

# SAGE-MPPI: Stagnation-Aware Geometry-Shaped Exploration for Sampling-Based Manipulation MPC

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## Abstract

Sampling-based model predictive control (MPC) is attractive for reactive manipulation because it can optimize nonlinear, nonconvex, and nonsmooth objectives under receding-horizon feedback without requiring analytic derivatives of the full objective [2], [5], [22]. Yet, in cluttered manipulator scenes, practical MPPI controllers still exhibit a persistent failure mode: after warm-starting, the sampling distribution may collapse around a locally feasible but globally unproductive basin and repeatedly resample the same low-progress region. This issue is especially pronounced in joint-space manipulation, where task objectives, self-collision avoidance, state bounds, and environment-clearance terms interact over short horizons.

We propose *SAGE-MPPI*, a lightweight proposal-shaping strategy for manipulator MPC. The method combines three simple ingredients: a stage-scaled covariance schedule, a normalized safe-elite anisotropic covariance, and a stagnation-triggered covariance amplification rule. The resulting controller preserves the standard MPPI weighted update and is designed to integrate into the current UR7e control stack by modifying only the sampling logic. The paper formulates each component in a code-consistent way and develops an experimental protocol for comparing the proposed method against fixed-covariance and diffusion-annealed baselines.

## 1 Introduction

Sampling-based receding-horizon control has become an important paradigm for robotics because it can optimize under nonlinear dynamics and arbitrary cost compositions while remaining compatible with parallel rollout evaluation [5], [22]. Among these methods, model predictive path integral (MPPI) control is particularly attractive: it transforms stochastic optimal control into a sampling-and-reweighting procedure that is simple, derivative-free at the system level, and empirically effective across manipulation and locomotion tasks [2], [22], [23]. For high-dimensional manipulation, STORM showed that joint-space sampling-based MPC can run at practical rates while directly accounting for task-space objectives, state constraints, self-collision, and environment collision [2]. More recently, diffusion-style annealing strategies have been proposed to improve the exploration–convergence trade-off of MPPI by varying sampling covariance across optimization stage and horizon index [14], [23].

Despite these advances, local-minimum behavior remains a central obstacle to reliable reactive manipulation. In cluttered scenes, the controller must search with a proposal distribution that is broad enough to discover alternate basins, directional enough to exploit manipulator-specific geometry, and stable enough to avoid unreliable early covariance estimates. In practice, however, a fixed isotropic covariance tends to underexplore narrow passages, while an aggressively adapted anisotropic covariance may overfit noisy early elites. Diffusion-style schedules alleviate part of this problem by imposing coarse-to-fine exploration, but isotropic annealing alone does not encode which joint-space directions are actually useful for safe progress. The result is a common failure mode:

the controller warm-starts around a basin that is locally admissible, terminal progress becomes negligible, and the sample budget is repeatedly reinvested in the same region.

This paper addresses that problem in the concrete setting of an existing UR7e STORM/MPPI codebase. The baseline branch contains a STORM-style UR7e controller in Gazebo, while the current whole-control branch already supports a distance-field collision cost and a diffusion-annealed controller variant. These two baselines motivate a central question: *how should one adapt the proposal covariance so that sampling-based MPC remains exploratory enough to escape clutter-induced local minima, yet simple enough to integrate into an existing manipulator control stack?*

Our answer is **SAGE-MPPI** (Stagnation-Aware Geometry-shaped Exploration MPPI). Rather than treating covariance as a fixed hyperparameter, we treat it as a structured object with separate scale and shape. Scale is governed by a compact exponential schedule across optimization stage and horizon position, while shape is estimated only from low-cost rollouts that also maintain positive clearance. When task progress stalls, the covariance magnitude is amplified by a single proportional factor, temporarily restoring exploratory capacity without otherwise changing the control law.

The contribution is therefore not another large architectural layer, but a compact covariance design tailored to manipulator MPC. Relative to existing approaches, the proposed method has three distinguishing properties: (i) it uses a *stage-scaled* covariance that preserves coarse-to-fine exploration with a lightweight parameterization; (ii) it constructs an anisotropic covariance from a mathematically explicit *safe elite set*, thereby using low-cost and positive-clearance evidence simultaneously; and (iii) it introduces a *stagnation-triggered amplification* rule that can be implemented with minimal changes to the existing controller.

## 2 Related Work

### 2.1 Offline Path Planning

Offline motion planning methods decouple path generation from execution and remain the classical reference point for high-dimensional robot planning. Sampling-based planners such as probabilistic roadmaps and rapidly-exploring random trees provide broad configuration-space coverage and probabilistic completeness, which makes them effective for difficult global search problems [7], [8], [12]. A complementary line of work formulates path generation as continuous optimization. Representative methods include CHOMP, STOMP, and TrajOpt, which optimize collision-aware trajectories by exploiting smooth surrogate objectives or sequential convexification [6], [15], [18]. More recent systems, such as GPU-accelerated trajectory generation frameworks, push offline planning toward higher throughput and better practical deployability [19].

These methods establish important algorithmic baselines, but they are not the primary object of this paper. Their key limitation in our setting is structural: they separate the planning problem from the feedback problem. For rapidly changing manipulator scenes, the difficulty is not only to find a collision-free path, but to continually reshape the search distribution online as the controller replans from new states.

### 2.2 Receding Horizon Path Planning

Receding-horizon methods repeatedly solve finite-horizon subproblems online and therefore occupy the middle ground between offline planning and purely local reaction. In robotics, this class includes nonlinear MPC, trajectory optimization with repeated horizon shifts, and stochastic optimal control methods based on path integral sampling [5], [22]. Sampling-based MPC is particularly relevant because it can handle complex cost functions and non-smooth events without requiring analytic

derivatives of the full objective. STORM demonstrated that this idea can be carried into high-dimensional joint-space manipulation, where task objectives, state constraints, and collision terms are optimized within a unified rollout-based controller [2].

A recent line of work has focused specifically on proposal design. Diffusion-style annealing improves the exploration–convergence trade-off by varying the sampling variance across inner optimization stage and horizon index [14], [23]. Other methods broaden the proposal family through biased importance sampling, learned sampling distributions, or multi-modal proposal structures [1], [17], [20], [25]. These works make clear that covariance design is not a secondary implementation choice but a central algorithmic component of sampling-based MPC.

## 2.3 Reactive Planning

Reactive planning emphasizes immediate adaptation to local geometry and disturbances, often prioritizing responsiveness over long-horizon global optimality. Classical artificial potential fields and navigation functions established the control-theoretic intuition that motion generation can be cast as a closed-loop flow on a potential landscape [9], [16]. However, they also exposed the fundamental difficulty of local minima. This theme continues to motivate modern reactive methods for manipulation.

One branch of recent work develops hierarchical reactive systems that combine fast local execution with a higher-level predictive or planning component. RAMP, for example, combines an MPPI-based trajectory generator with a vector-field follower, thereby balancing global trajectory generation with local reactive safety [21]. A second branch focuses on dynamical-system formulations and their interaction with optimization. Koptev *et al.* combine dynamical-system modulation with sampling-based MPC to overcome local minima induced by obstacle geometry in joint space [11]. Closely related work on neural joint-space signed distance functions further shows that geometry-aware local control can be made computationally practical for manipulator settings [10]. A third branch studies reactive local motion generation more broadly, including velocity-level redundancy-resolution controllers and fast local planners for constrained tasks [3], [4], [13], [24], [26].

These methods are highly relevant because they frame local-minimum avoidance as a geometric and feedback problem rather than a purely global planning problem. Our setting differs, however, in one important respect: the controller itself is already a sampling-based receding-horizon optimizer. The question is therefore not how to add an external reactive layer, but how to make the proposal distribution of MPPI itself more robust to clutter-induced local minima.

## 3 Method

We now formulate a lightweight covariance design for MPPI. The objective is to preserve the simplicity of the original algorithm while modifying the sampling rule so that it becomes stage-aware, geometry-aware, and stagnation-aware.

### 3.1 Problem Formulation

Consider a manipulator with state  $x_t \in \mathbb{R}^{n_x}$  and control sequence

$$U = (u_0, \dots, u_{H-1}), \quad u_h \in \mathbb{R}^d, \quad (1)$$

where  $d$  is the control dimension, i.e., the number of commanded joint-space inputs at each horizon step. Given the current state  $x_t$ , MPPI samples  $N$  control sequences  $\{U^{(n)}\}_{n=1}^N$  from a proposal

distribution, rolls them forward under the system dynamics, and assigns each rollout a total cost  $J^{(n)}$ . The standard MPPI weights are

$$w_n = \frac{\exp(-J^{(n)}/\lambda)}{\sum_{m=1}^N \exp(-J^{(m)}/\lambda)}, \quad (2)$$

with temperature  $\lambda > 0$ . The mean sequence is then updated by the weighted average

$$\mu_{k+1,h} = \sum_{n=1}^N w_n u_{k,h}^{(n)}, \quad h = 0, \dots, H-1, \quad (3)$$

where  $k$  indexes the inner MPPI iteration.

The present work does not alter Eqs. (2) and (3). Instead, it modifies the proposal covariance used to generate the rollouts.

### 3.2 Stage-Scaled Proposal

We factorize the proposal covariance as

$$\Sigma_{k,h} = s_{k,h} C_{k,h}, \quad (4)$$

where  $s_{k,h} > 0$  is a scalar scale and  $C_{k,h} \in \mathbb{R}^{d \times d}$  is a normalized shape matrix. The nominal scale follows the compact exponential schedule

$$s_{k,h} = \sigma_0 \exp\left(\sigma_1 \frac{h-H}{H} - \sigma_2 \frac{k}{K}\right), \quad (5)$$

where  $\sigma_0$  is the base covariance magnitude and  $\sigma_1, \sigma_2 > 0$  control horizon-wise broadening and iteration-wise cooling, respectively. This form preserves the intended diffusion-style behavior while remaining algebraically simple.

### 3.3 Safe-Elite Anisotropic Covariance

The scale in Eq. (5) controls *how much* to explore but not *in which directions* to explore. To capture directional structure, we estimate a covariance from safe elite rollouts.

For rollout  $n$ , let  $\delta^{(n)}$  denote its minimum signed safety margin over the horizon,

$$\delta^{(n)} = \min_{0 \leq h < H} \min_{\ell \in \mathcal{L}} \min_{s \in \mathcal{S}_\ell} d(p_{h,\ell,s}^{(n)}), \quad (6)$$

where  $p_{h,\ell,s}^{(n)}$  is the center of robot sphere  $s$  attached to link  $\ell$  at rollout step  $h$ , and  $d(\cdot)$  is the signed clearance returned by the environment distance field after subtracting the sphere radius and the controller safety margin. In the current codebase, this quantity is directly available by recording the minimum pre-clamp signed clearance along each rollout.

Let  $Q_\eta(\{J^{(m)}\}_{m=1}^N)$  denote the empirical  $\eta$ -quantile of rollout costs, namely the smallest scalar  $q$  such that at least an  $\eta$  fraction of the sampled rollouts satisfy  $J^{(m)} \leq q$ . In code, this is obtained by sorting the batch costs and selecting the element at index  $\lceil \eta N \rceil$ . We then define the safe elite set as

$$\mathcal{E}_t = \left\{ n \in \{1, \dots, N\} \mid J^{(n)} \leq Q_\eta(\{J^{(m)}\}_{m=1}^N), \delta^{(n)} > 0 \right\}. \quad (7)$$

This definition is intentionally implementation-oriented: a rollout belongs to  $\mathcal{E}_t$  only if it is both low-cost relative to the current batch and strictly clearance-feasible.

If  $\mathcal{E}_t = \emptyset$ , we skip anisotropic adaptation and set  $C_{k,h} = \mathbf{I}$ . Otherwise, we renormalize the MPPI weights over  $\mathcal{E}_t$ ,

$$\bar{w}_n = \frac{w_n \mathbb{I}[n \in \mathcal{E}_t]}{\sum_{m=1}^N w_m \mathbb{I}[m \in \mathcal{E}_t]}, \quad (8)$$

and compute the stepwise empirical covariance

$$\hat{C}_{k,h} = \sum_{n=1}^N \bar{w}_n (u_{k,h}^{(n)} - \mu_{k,h})(u_{k,h}^{(n)} - \mu_{k,h})^\top. \quad (9)$$

To separate covariance *shape* from global *scale*, we normalize it by its average eigenvalue,

$$\tilde{C}_{k,h} = \frac{\hat{C}_{k,h}}{\text{tr}(\hat{C}_{k,h})/d}, \quad (10)$$

where  $d$  is again the control dimension. If  $\hat{C}_{k,h} = 0$ , which in code means its trace is numerically zero, we set  $\tilde{C}_{k,h} = \mathbf{I}$ . Thus, the magnitude of exploration remains governed by [Eq. \(5\)](#), while [Eq. \(10\)](#) contributes only directional structure.

To avoid trusting noisy covariance estimates too early, we blend isotropic and anisotropic proposals using

$$\rho_k = \frac{k}{K} \sum_{n=1}^N w_n \mathbb{I}[n \in \mathcal{E}_t], \quad (11)$$

and define

$$C_{k,h} = (1 - \rho_k)\mathbf{I} + \rho_k \tilde{C}_{k,h}. \quad (12)$$

Since the global MPPI weights satisfy  $\sum_n w_n = 1$ , the quantity  $\sum_n w_n \mathbb{I}[n \in \mathcal{E}_t]$  is precisely the weight mass carried by the safe elite set. Consequently, the controller starts from broader isotropic exploration and transitions toward a geometry-shaped proposal only after enough evidence has accumulated.

### 3.4 Stagnation-Triggered Amplification

The previous section determines the nominal covariance schedule. We next introduce a minimal mechanism for restoring exploration when the controller stalls.

Let  $q$  denote the joint-position component of the state  $x$ , and let  $p_{\text{ee}}(q)$  be the forward-kinematics map that returns the end-effector position. For a reaching task with target position  $p_g$ , we define

$$g(x) = \|p_{\text{ee}}(q) - p_g\|_2. \quad (13)$$

This quantity is directly computable in the current controller because both the robot kinematics and the Cartesian target are already available during rollout evaluation. The executed one-step progress is then

$$\Delta_t^{\text{goal}} = g(x_{t-1}) - g(x_t), \quad (14)$$

and the binary stagnation indicator is

$$z_t = \mathbb{I} \left[ \Delta_t^{\text{goal}} < \tau_p \right], \quad (15)$$

where  $\tau_p > 0$  is a small progress threshold. When stagnation is detected, we amplify the covariance scale according to

$$s_{k,h} \leftarrow (1 + \alpha z_t) s_{k,h}, \quad (16)$$

where  $\alpha > 0$  is a proportional gain controlling the strength of re-exploration. This rule deliberately leaves the anisotropic shape unchanged; it only broadens the search.

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**Algorithm 1** SAGE-MPPI at control step  $t$ 

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**Require:** Current state  $x_t$ , warm-start mean  $\mu_{0:H-1}$

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1: for  $k = 1$  to  $K$  do
2:   for  $h = 0$  to  $H - 1$  do
3:     Compute  $s_{k,h}$  from Eq. (5)
4:   end for
5:   if  $z_t = 1$  then
6:     Amplify  $s_{k,h}$  using Eq. (16)
7:   end if
8:   Sample  $N$  rollouts from  $\mathcal{N}(\mu_{k,h}, s_{k,h}C_{k,h})$ 
9:   Roll out dynamics and evaluate total costs  $\{J^{(n)}\}_{n=1}^N$ 
10:  Compute MPPI weights  $\{w_n\}_{n=1}^N$  using Eq. (2)
11:  Compute safety margins  $\{\delta^{(n)}\}_{n=1}^N$  using Eq. (6)
12:  Form  $\mathcal{E}_t$  from Eq. (7)
13:  if  $\mathcal{E}_t \neq \emptyset$  then
14:    Compute  $\bar{w}_n$  via Eq. (8)
15:    for  $h = 0$  to  $H - 1$  do
16:      Compute  $\hat{C}_{k,h}$  via Eq. (9)
17:      Normalize  $\hat{C}_{k,h}$  via Eq. (10)
18:      Compute  $\rho_k$  via Eq. (11)
19:      Set  $C_{k,h}$  via Eq. (12)
20:    end for
21:  else
22:    Set  $C_{k,h} = \mathbf{I}$  for all  $h$ 
23:  end if
24:  Update the mean sequence using Eq. (3)
25: end for
26: Execute the first control and shift the remaining sequence forward
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### 3.5 Final Proposal and Algorithm

Combining the previous components, the final proposal at inner iteration  $k$  and horizon step  $h$  is

$$u_{k,h} \sim \mathcal{N}(\mu_{k,h}, \Sigma_{k,h}), \quad \Sigma_{k,h} = s_{k,h}C_{k,h}, \quad (17)$$

with  $s_{k,h}$  given by Eqs. (5) and (16) and  $C_{k,h}$  given by Eq. (12). The resulting controller is summarized in Algorithm 1.

### 3.6 Implementation in the Current UR7e Stack

The proposed method is designed to fit the current project structure. The existing UR7e Gazebo baseline already provides a STORM-style controller wrapper, and the whole-control branch already exposes rollout evaluation with a distance-field collision cost. Therefore, SAGE-MPPI requires only three implementation changes: (i) logging the per-rollout minimum signed safety margin in Eq. (6); (ii) building the safe elite set and its normalized covariance inside the controller loop; and (iii) replacing the fixed or diffusion-style covariance with Eq. (17). No changes are required to the task specification, rollout model, command interface, or collision representation.

## 4 Experimental Setup

This section describes the experimental protocol to be implemented; no numerical results are reported here.

**Baselines.** We recommend four controllers: (i) the original STORM-style MPPI controller in the main branch [2]; (ii) the whole-control branch with fixed-covariance MPPI; (iii) the diffusion-annealed baseline inspired by DIAL-MPC [23]; and (iv) the proposed SAGE-MPPI.

**Scenes.** Experiments should begin with the existing reaching and tall-obstacle scenes already supported in the repository. Additional scenes can be generated by varying target position, obstacle height, and corridor width while preserving the same UR7e kinematic model and collision representation.

**Metrics.** The main metrics are: success rate, final end-effector position error, minimum rollout clearance, time-to-goal, average control frequency, and stagnation incidence. A useful additional metric is *recovery rate*: the fraction of stalled episodes that later regain positive progress according to Eq. (14).

**Ablations.** At minimum, the following ablations should be included: (a) stage scale only; (b) stage scale plus normalized anisotropic covariance; (c) stage scale plus amplification; and (d) full SAGE-MPPI. This isolates the contributions of staged exploration, safe-elite geometry shaping, and stagnation-triggered recovery.

**Compute Budget.** For fair comparison, all controllers should use the same horizon length, number of particles, control frequency, and GPU device. This is especially important because the proposed method changes proposal design rather than rollout complexity.

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